calm_weighted_matched_legacy

equity strategy | family: equity-volatility | status: reference | derived from `calm_weighted_matched`

What this strategy does

Calm Weighted resized so its ups and downs match Balanced Leverage closely (sized on the pre-fix book; same volatility to within 0.1 point after the fixes) -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. PEAD is a stronger, cleaner signal in low-vol stocks; size positions by inverse risk.

Design notes (from the strategy's source file).

Calm Weighted -- Balanced Leverage, with bigger bets on calmer stocks (sized by inverse idiosyncratic volatility).

Within every (quarter, side, sector, size) cell, each event's weight is scaled by 1/ivol, so calm stocks get more capital than wild ones. Rationale: the earnings drift is worth more per unit of risk in low-vol names (the SUE x vol interaction is t = -2.4 to -2.7 in Fama-MacBeth), and the engine's cost/liquidity model charges the same bps regardless of vol. Raises Sharpe with no fitted parameter, in all three sub-eras (corrected run: Balanced Leverage 1.09 -> Calm Weighted 1.24; pre-fix 1.30 -> 1.47).

`CalmWeightedMatched` re-sizes it to Balanced Leverage's exact volatility to answer "at the SAME risk, how much more return" (corrected run: 5.83% vs 5.38%; pre-fix 7.31% vs 6.49%).

parameter	value
base_unit_fraction	0.003468808347716693
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	calm weighted matched legacy	market (CRSP VW)
Annualized return	7.31%	9.94%
Annualized volatility	4.94%	19.35%
Sharpe (NAV-pct)	1.45	0.59
Sortino	2.15	
Max drawdown	-17.0%	-55.5%
Calmar	0.43	
Final NAV ($10M start)	$39,500,315	
Worst day	-3.35%	
% positive days	52.4%	

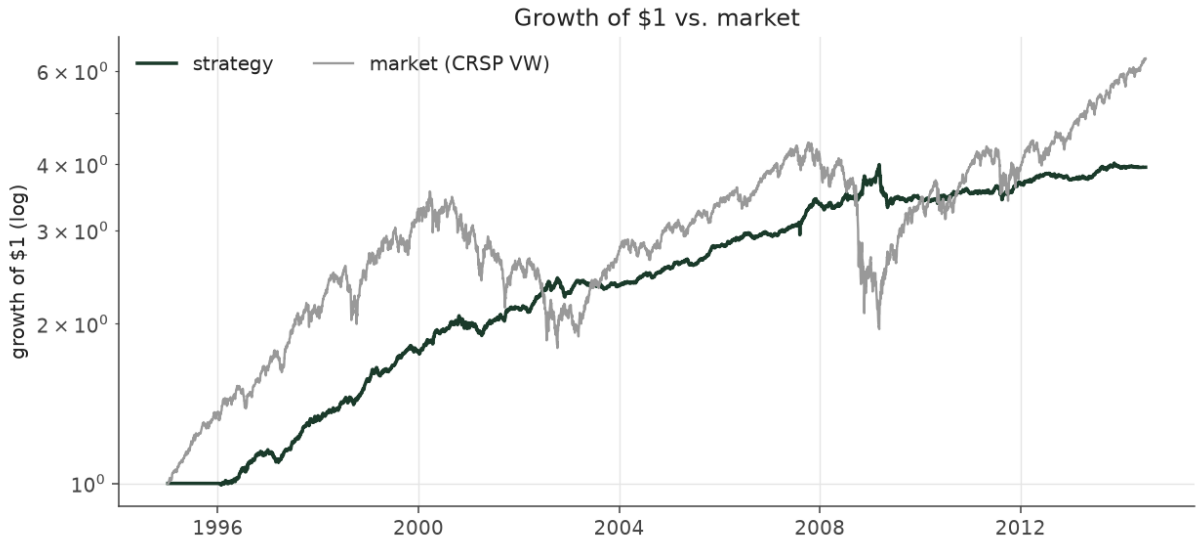
Sharpe 1996-2001	2.39	
Sharpe 2002-2007	1.66	
Sharpe 2008-2013	0.52	

Correlation of daily returns with the market: **-0.00**.

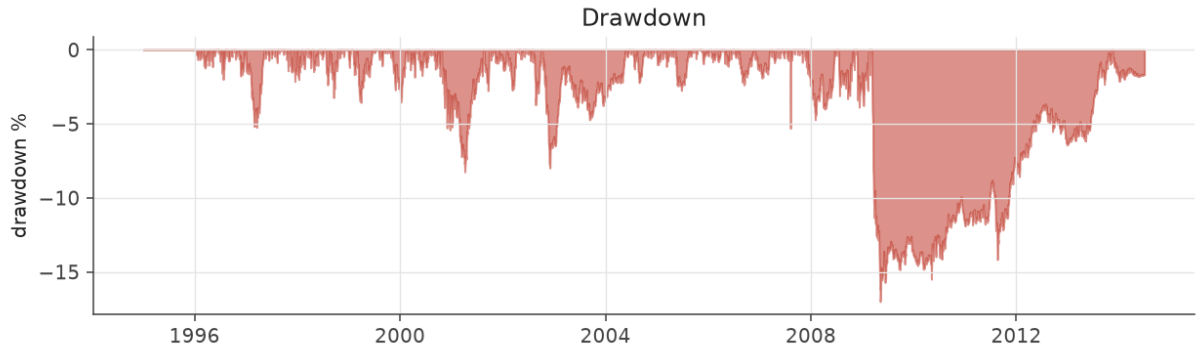
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.45

• best return among equity: `market_boost_max` (0.116); this one: 0.0731

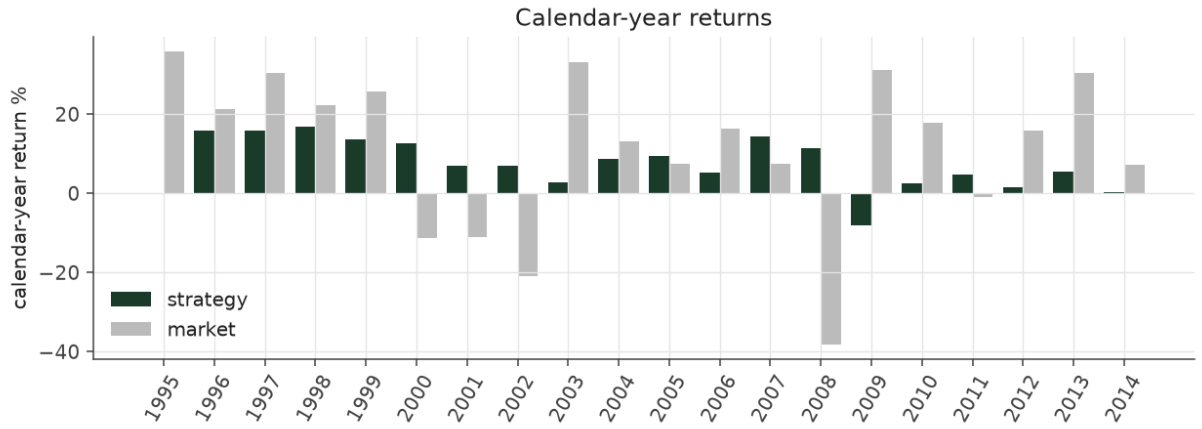
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.43



NAV (log scale)



Drawdown from running peak

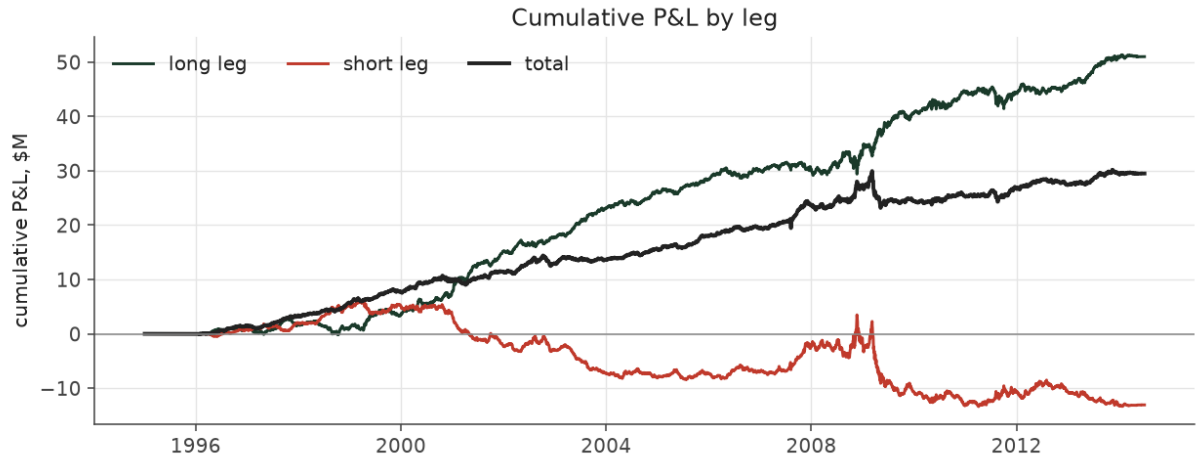


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	51.01	-13.08
P&L after costs ($M)	46.96	-17.46
of which common (average-event) component ($M)	33.29	-34.09

of which excess / PEAD component ($M)	17.26	21.34
excess minus costs ($M)	13.22	16.98
gross return per $ traded (bp)	242.65	-60.53
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.32	-0.23

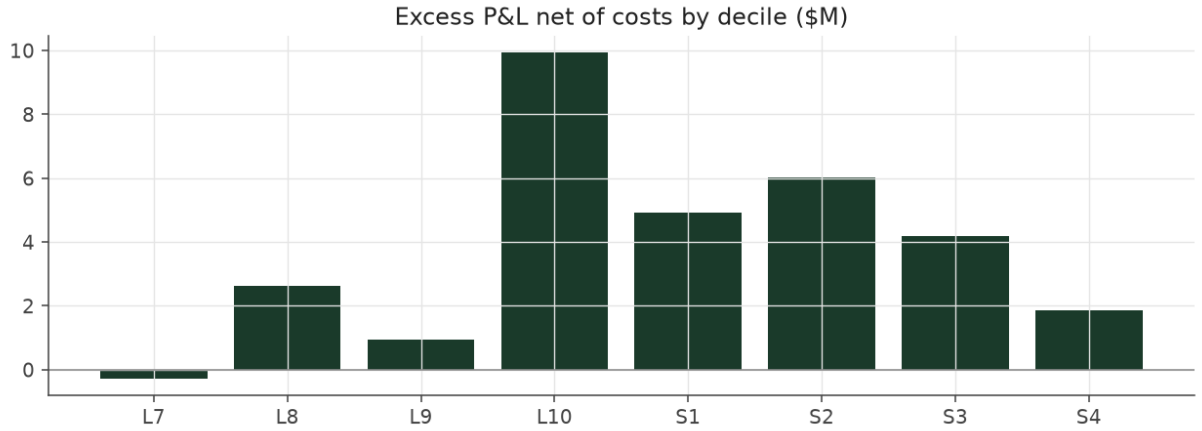
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-13.1M before costs) while the long leg earns +51.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 33.3M / 34.1M. Stripping that common component, the surprise signal itself earned +17.3M on the long side and +21.3M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.32 long / -0.23 short vs 1.45 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,017	187.53	3.24	0.07	0.36	-0.29	-1.0%	-15.53
long	8	16,581	496.58	10.87	3.58	0.96	2.61	8.7%	52.63
long	9	17,745	651.92	10.47	2.19	1.24	0.95	3.1%	14.58
long	10	17,173	766.04	22.39	11.42	1.48	9.94	32.9%	129.72
short	1	17,166	766.10	-6.38	6.53	1.61	4.92	16.3%	64.23
short	2	17,276	669.92	-5.27	7.35	1.34	6.01	19.9%	89.73
short	3	15,967	514.75	-4.22	5.20	1.02	4.17	13.8%	81.01
short	4	9,161	209.99	-1.58	2.26	0.40	1.86	6.2%	88.73



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,194	334.35	5.62	7.85	1.28	6.57	21.7%	196.48
2	22,880	766.10	6.70	9.29	2.16	7.14	23.6%	93.14

3	25,498	978.87	7.90	9.64	2.06	7.58	25.1%	77.41
4	27,576	1,082.62	5.42	6.90	1.71	5.18	17.2%	47.89
5	28,938	1,100.90	3.87	4.92	1.21	3.71	12.3%	33.68

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,631	247.89	5.24	6.97	1.24	5.73	19.0%	231.15
2	23,084	835.03	6.57	10.23	2.50	7.72	25.6%	92.50
3	25,848	1,073.98	8.16	10.79	2.15	8.64	28.6%	80.44
4	27,653	1,066.69	5.75	6.36	1.49	4.87	16.1%	45.63
5	28,865	1,038.89	3.74	4.21	1.04	3.17	10.5%	30.48

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,837	1,657.83	0.68	9.89	2.91	6.98	23.1%	42.11
41-60d	25,647	973.48	6.02	5.96	1.96	4.00	13.2%	41.07
61-90d	35,313	1,222.87	8.62	10.31	2.55	7.76	25.7%	63.47
>90d	12,289	408.66	14.18	12.45	1.01	11.44	37.9%	279.81

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	809.38	11.18	12.87	1.57	11.30	37.4%	139.60
2002-07	39,913	1,427.49	12.27	14.90	2.77	12.14	40.2%	85.05
2008-13	45,278	2,025.98	6.06	10.82	4.09	6.74	22.3%	33.24

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,085	427.07	2.45	3.36	0.90	2.46	8.1%	57.49
Chems	2,588	279.38	1.84	2.21	0.48	1.73	5.7%	61.84
Durbl	2,843	300.63	2.59	3.18	0.53	2.64	8.7%	87.90
Enrgy	6,194	342.20	4.51	5.28	0.63	4.65	15.4%	135.97
Hlth	11,685	411.81	3.00	3.81	0.86	2.96	9.8%	71.83
Manuf	12,033	383.08	2.04	2.92	0.78	2.14	7.1%	55.87
Money	17,569	375.06	3.08	4.09	0.83	3.26	10.8%	86.92
NoDur	5,080	341.14	1.72	2.33	0.63	1.70	5.6%	49.75
Other	18,372	410.74	3.06	4.12	0.91	3.21	10.6%	78.04
Shops	11,984	398.66	3.75	4.54	0.81	3.73	12.3%	93.56

Telcm	3,981	327.17	0.81	1.42	0.58	0.84	2.8%	25.75
Utils	4,672	265.90	0.66	1.34	0.48	0.86	2.9%	32.45

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,494	1,057.22	4.66	7.73	1.92	5.82	19.3%	55.04
low vol	40,087	1,759.86	13.57	17.57	3.70	13.87	45.9%	78.83
mid vol	37,893	1,388.11	11.06	13.11	2.69	10.43	34.5%	75.12

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	7.31%	-0.00%
0.5%	2.2	7.00%	0.31%
1.0%	4.3	6.67%	0.64%
2.0%	8.7	5.95%	1.36%
5.0%	21.7	3.01%	4.30%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.